

Machine Learning Data Engineer with hands-on experience building scalable data pipelines, feature engineering workflows, and end-to-end ML systems. Proficient in Python and modern data tooling (Spark, Databricks, Airflow), with experience productionizing ML models, developing AI-driven automation, and owning the full ML data lifecycle — from data ingestion and feature engineering through deployment, monitoring, and continuous optimization — in production-like environments.

SKILLS

Programming & Querying	Python, Java, SQL, C/C++, Bash
Data Engineering	Apache Spark, Airflow, Databricks, ETL Pipelines, Feature Stores, Snowflake, MLflow, Docker
ML & AI	PyTorch, scikit-learn, XGBoost, Feature Engineering, Model Deployment & Monitoring, AI Agents, LangChain
Cloud & DevOps	AWS (SageMaker, Glue), GitHub Actions, CI/CD, Git/GitHub, Linux

EDUCATION

Master of Science in Applied Computing (CGPA 4.0/4), University of Toronto Artificial Intelligence Focus	September 2024 - December 2025 Toronto, ON, Canada
Honors Bachelor of Science with High Distinction (CGPA 3.94/4), University of Toronto Specialist in Computer Science, Major in Mathematics, Minor in Statistics	September 2018 - April 2024 Toronto, ON, Canada

TECHNICAL EXPERIENCE

ML Data Engineer Intern Python, PyTorch, XGBoost, Databricks, Spark, Docker, NVIDIA Triton Magna New Mobility	May 2025 – December 2025 Toronto, ON, Canada
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- Designed and built a **scalable data ingestion and transformation pipeline** for ultra-high-resolution industrial images, orchestrating patch-level preprocessing and window-level prediction to construct **ML-ready datasets** under production compute constraints.
- Engineered **feature pipelines and data augmentation strategies** that drove measurable improvements in small-defect detection (**+0.19 IoU, +0.30 Precision, +0.22 Recall**) through systematic experimentation across architectural variants and pipeline configurations.
- Owned the **end-to-end ML data lifecycle** for a leather defect analysis system — from raw data ingestion and feature engineering through multi-stage model integration (**segmentation → crop extraction → XGBoost classification**) and structured evaluation — enabling interpretable severity modeling at scale.
- Built and deployed **production-grade ML pipelines** on **Databricks and Docker**, integrating **NVIDIA Triton** for scalable inference; implemented deployment-aware workflows supporting **reproducible experimentation, monitoring, and continuous optimization**.

Software Engineer Intern C++, Python, ROS2 Foxy, Linux, Git Magna Electronics	May 2022 - June 2023 Brampton, ON, Canada
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- Improved **data quality and reliability** for vision-based sensing through intrinsic and extrinsic camera calibration, ensuring accurate sensor alignment and validated data integrity for downstream perception pipelines and model evaluation.
- Enhanced **data pipeline robustness** by debugging and extending C++/Python processing modules, reducing failure rates under real-world conditions and maintaining consistent data throughput for production systems.
- Applied **CI/CD and DevOps practices** — including version-controlled deployments, automated testing, and code review workflows — to support reliable **ROS2 Foxy platform integration on Linux** and ensure reproducible, deployment-ready data infrastructure.

PROJECTS

INTR: Interpretable Transformer for Image Classification Python, PyTorch, git	September 2024 - December 2024
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- Conducted a **large-scale evaluation study** of INTR across **14 fine-grained datasets** (8 original + 6 extended), combining **accuracy benchmarking** with attention-based diagnostics.
- Analyzed **generalization behavior** by benchmarking INTR against **ResNet-50**, identifying dataset-dependent performance shifts, including up to **+9.2% accuracy gain** on Fish and up to **-12% degradation** on CUB under complex background conditions.
- Performed **interpretability-driven analysis** using **class-specific cross-attention patterns** to attribute model predictions to discriminative regions, supporting error analysis in fine-grained classification.

Artwork Restoration with Generative Image Inpainting Python, PyTorch, git	September 2024 - December 2024
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- Designed a **controlled benchmarking study** to evaluate **3 generative inpainting models** (CO-MOD-GAN, ICT, LaMa) on the **WikiArt dataset**, training on **20k images** and testing on a **5k-image holdout set** with **5 degradation mask types**.

- Quantified restoration performance using **MSE, MS-SSIM, and FID**, revealing clear model trade-offs: LaMa achieved the best **pixel-level and structural fidelity** (MSE 0.01002, MS-SSIM 0.79205), while Co-Mod-GAN produced the most realistic global distributions (**FID 112.155**).
- Conducted **qualitative error analysis** across diverse art styles to identify **systematic failure modes**, including facial distortion, fine-detail degradation, and performance collapse under full occlusion.